

[WORKING TITLE: Composing a Conservative Memory Primitive: a Degree-Bounded Interference Firewall and a Priced Channel Through It]

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Abstract

We ask what happens to a conservative, physics-structured memory *primitive* when it is scaled from one unit to many, and answer it for a lattice of damped-symplectic recurrent units advanced by a single **joint Hamiltonian** with **sparse position-only coupling**. The construction inherits its per-unit guarantees by composition — joint conformal symplecticity, an exact vanishing-coupling reduction, a per-unit Noether budget, and a mass-anisotropic causal cap all hold on the joint state to machine precision, and the communication cost predicted for a shared channel (relative-mode retention $\propto \kappa^{-1}$, synchronization $\propto \kappa^{-1/2}$) is verified on designed lattices to fitted slopes $-0.986/-0.499$. Three results are then measured on lattices with learned MLP potentials. **(1) The interference firewall is measured, and it is *degree-bounded*, not *width-dependent*:** a localized cross-unit contrastive-divergence update displaces a non-adjacent unit's basin by **exactly zero** (0 of 4,656 off-graph entries nonzero over 72 runs; $R \equiv 0$ off the coupling graph in all three topologies tested), so per-unit received interference obeys the exact identity $S = \deg \cdot \bar{R}_{\text{edge}}$ (to six decimals) with a per-edge leak $\bar{R}_{\text{edge}} \approx 6.5 \times 10^{-5}$ of the intended change; at fixed degree S is statistically flat in N (ring, $\deg \equiv 2$: slope $b = +0.07 \pm 0.18$, $p = 0.39$) and a degree-4 circulant is $2\times$ the ring. A same-family, same-width shared-potential monolith has no such confinement (the arms are width-matched, not parameter-matched; §5 and the capacity controls below): it displaces *every* other unit by ≈ 0.19 with no spatial structure and grows as $N^{1.18 \pm 0.17}$, so that by $N \approx 6$ its aggregate cross-unit force perturbation exceeds a unit's own update magnitude — **degree-bounded vs width-linear**, slopes diverging (Welch $p = 3.3 \times 10^{-4}$), separation $\times 20,649$ at $N = 16$ ($N \leq 16$, chain/ring/circulant-4, 8–12 seeds, 2-dim units, MLP potentials, $\kappa = 0.05$, measured at initialization, laptop-CPU). The firewall is bought by **parameter separation, not block structure**: a block-diagonal single potential with *shared* parameters still interferes (a permutation-equivariant deep-sets potential worst of all, per-pair $\bar{R} = 0.97$), while one with *separate* per-block parameters gives $S \equiv 0$ exactly; the modular lattice is that firewall plus an $O(\kappa^2)$ graph-local leak, bought in exchange for communication. This is the paper's thesis in one line: **modularity is the firewall, and the physics is the priced channel one gets to keep while having it** (§3.2(iii), §3.3) — a single joint symplectic Hamiltonian under which units talk at a measured, graph-local cost. **(2) That priced channel obeys a predictive price list, and the list is not an $N = 2$ artifact** (trained lattices, $N \leq 16$): an effective coupling curvature κ_{eff} read *blind* from the learned coupling potential predicts recall-horizon ranking (Spearman 1.0, $n = 5$, exact $p = 0.008$) and synchronization time to $\leq 7.5\%$ of the registered prediction across a $91\times$ decade on trained 2-unit lattices (3 seeds), and the underlying price law ($\text{sync} \propto \kappa_{\text{eff}}^{-1/2}$, $n_{1/2} \propto \kappa_{\text{eff}}^{-1}$) holds **flat in N** at $N \in \{2, 4, 8, 16\}$, both chain and ring, on $U(1)$ -preserving (**channel_spring**) trained lattices of designed $SO(2)$ units (5 seeds; sync exponent -0.49 ± 0.02 , $n_{1/2}$ exponent -0.91 ± 0.03 , $R^2 \geq 0.998$), all predictions registered before measurement — so the physics-specific priced channel spans the same $N \leq 16$ range as the firewall, provided the coupling preserves the $U(1)$

symmetry. **(3) Banding is a method, not an oracle gift** (trained 2-unit lattices, $16\times$ ratio, 5 seeds): a correct inertial-mass prior halves the held-out error, a wrong prior of identical spread costs up to $10.8\times$, and a cheap from-data spectral (FFT) selector recovers the oracle band exactly (MSE gap 0.000, 5/5 seeds) *when timescales are spectrally separable*. We close with the price of the physics prior, measured through the horizon: a param-matched unconstrained twin leads by 1–2 orders of magnitude to ≈ 500 steps, is overtaken at ≈ 700 , and diverges (MSE 196 at 5000 steps vs the single CLU unit’s bounded 0.20–0.23 plateau — this control is a single unit, $N = 1$, dim 4). As a systems corollary, the separable Hamiltonian makes the $\gamma = 0$ lattice exactly reversible, giving recompute-backwards BPTT at $O(1)$ -in- T activation memory ($946\times$ lower peak memory at $T = 1024, N = 2$) with float-indistinguishable gradients ($\leq 2.1 \times 10^{-6}$ relative, float32) — exact only along the conservative direction ($\gamma = 0$; $\gamma > 0$ gives a finite reversal horizon $\propto (1 - \gamma)^{-n}$), validated on *untrained* models, not yet wired into the shipped trainer. All experiments are laptop-CPU; designed-testbed numbers are labeled verification, learned-system numbers evidence, integrator/autodiff-graph numbers structural.

1 Introduction

A structure-preserving alternative to gated recurrence advances a latent state (q, p) by a **damped symplectic (velocity-Verlet) step of a learnable separable Hamiltonian** $H(q, p) = T(p) + V_\theta(q)$, with an optional per-step momentum damping $p \mapsto (1 - \gamma)p$ supplying controllable forgetting. Our reference unit is the **CLU (Causal Learning Unit)**, introduced as **CHLU in Jawahar & Pierini (2026)**; the exactly-solvable single-unit theory it obeys is developed in a companion note (Anonymous, 2026; hereafter *the theory note*). This paper is about the **scaling** question a single unit cannot answer: when you build a *network* of such units, do the per-unit guarantees survive, what does inter-unit communication cost, and can the per-unit timescales be chosen rather than gifted?

Our answer is a **CLU-Net**: a lattice under one joint Hamiltonian $H = \sum_i [T_i(p_i) + V_{\theta,i}(q_i)] + \sum_{(i,j) \in E} V_c(q_i, q_j)$ with sparse, **position-only** couplings on a graph E (so a joint symplectic step is well-defined and momentum coupling is impossible by construction). This is not a new cell but a *composition law*, and the thesis is threefold and, deliberately, ML-first: a joint Hamiltonian with sparse coupling **composes a conservative memory primitive** — with (i) a *measured* interference firewall that turns the catastrophic-interference risk of parameter sharing into a cost bounded by the coupling graph’s **coordination number rather than the network’s width**, (ii) a *predictive* communication price list read blind from learned curvature, and (iii) a *method* (not an oracle gift) for allocating per-unit timescales.

What the physics buys, and what it does not (stated up front). A reader will otherwise assemble this from a concession in §3.2(iii) and a scope line in §3.3, so we compose it here. The interference firewall — contribution (i), our headline — is **not bought by anything physics-specific**: *parameter isolation* buys it, and buys it strictly better. A single potential built from **disjoint per-unit parameter blocks** has *exactly zero* cross-unit interference (§3.2(iii)), where our lattice retains a small graph-local leak, and it needs no Hamiltonian, no symplectic structure and no coupling graph. **Modularity is the firewall; the physics is what one gets to keep while having it.** What a single joint symplectic Hamiltonian keeps — and a bag of disconnected potentials throws away — is a **priced channel** through the firewall: units that *communicate* under one conservative dynamics at a cost that is measured, graph-local and $O(\kappa^2)$ (contribution (ii), §3.3), together with reversibility (§3.5), conserved charges (§3.1), and the per-unit budget laws a shared Hamiltonian carries. That priced channel is our physics-specific claim: its predictive parity is demonstrated on **trained 2-unit lattices** (§3.3), and the underlying price law is **no longer an**

$N = 2$ result — it holds **flat in N** at $N \in \{2, 4, 8, 16\}$, both topologies, on $U(1)$ -preserving trained lattices of designed $SO(2)$ units (§3.3, Fig. 3), so the physics-specific contribution now **spans the same $N \leq 16$ range as the firewall**. The one qualifier is that the coupling must preserve the $U(1)$ symmetry: a symmetry-breaking random- W coupling clusters κ_{eff} so the exponent becomes unfittable — itself the resolution of Appendix C’s earlier “inconclusive on trained lattices,” not a limitation of the law.

Nomenclature (do not conflate). Two quantities both get called “mass” and run in opposite directions: the **inertial mass** M_i (the learned diagonal of T_i ; larger $M \Rightarrow$ slower) and the **spectral mass** μ_k (normal-mode frequency; larger $\mu \Rightarrow$ shorter memory). Per-unit timescales are set by the inertial-mass **band** $\{M_i\}$; retention statements use μ .

Contributions.

1. **The CLU-Net and its guarantees-by-composition (§3.1, verification).** Per-unit guarantees compose to machine precision — joint conformal symplecticity ($< 10^{-12}$, $x64$), an *exact* bit-level vanishing-coupling reduction, the joint Noether decay $(1 - \gamma)^n$, a mass-anisotropic cap $v_i^{\text{max}} = c/\sqrt{M_i}$ — and the price law is verified on designed lattices (slopes $-0.986/-0.499$ vs $-1/-\frac{1}{2}$).
2. **The interference firewall, measured — and it is *degree-bounded*, not width-dependent (§3.2, headline evidence).** Cross-unit CD basin displacement is **exactly zero off the coupling graph** (0/4,656 entries over 72 runs), giving the exact identity $S = \text{deg} \cdot \bar{R}_{\text{edge}}$; at fixed degree S is statistically flat in N (ring, $b = +0.07 \pm 0.18$), while the shared-potential monolith is **width-linear** ($N^{1.18 \pm 0.17}$; slopes diverge, Welch $p = 3.3 \times 10^{-4}$) — measured at $N \leq 16$ on chain/ring/circulant-4 lattices of 2-dim units with MLP potentials, $\kappa = 0.05$, 8–12 seeds, at initialization, laptop-CPU. The leak is $\propto \kappa^2$ and mass-independent, and the firewall is bought by **parameter separation, not block structure**.
3. **The price list is predictive, and it scales (§3.3).** *Evidence:* a coupling curvature read blind predicts recall-horizon ranking (Spearman 1.0) and sync (to $\leq 7.5\%$ of the registered prediction) across a $91 \times$ decade, registered before measurement (2-unit trained lattices, 3 seeds). *Verification (across composition):* the underlying price law holds **flat in N to $N \leq 16$** (sync exponent -0.49 ± 0.02 , $n_{1/2}$ exponent -0.91 ± 0.03 , $R^2 \geq 0.998$) on $U(1)$ -preserving **channel_spring** trained lattices of designed $SO(2)$ units (both topologies, 5 seeds, pre-registered).
4. **Banding as a method (§3.4, evidence).** A correct band halves held-out error; a wrong band of equal spread costs up to $10.8 \times$; a from-data spectral selector recovers the oracle band exactly when timescales are spectrally separable; a mass-lr is a partial optimizer-native inducer (ordering yes, magnitude no).
5. **Reversible $O(1)$ -memory training (§3.5, structural).** At $\gamma = 0$ the separable-Hamiltonian lattice admits exact recompute-backwards BPTT: gradients match stored-activation BPTT to $\leq 2.1 \times 10^{-6}$ (float32) at peak activation memory $O(1)$ -in- T vs $O(T)$ ($946 \times$ at $T = 1024$, $N = 2$; CPU/small- D), exact only at $\gamma = 0$ ($\gamma > 0$: finite reversal horizon $\propto (1 - \gamma)^{-n}$), and validated on *untrained* models — the gradient mechanics only, not yet wired into the shipped **train_chlu**.

6. The price of the physics prior, measured through the horizon (§3.6, evidence).

The fit gap decomposes into a dominant *contraction-forbidden* term and a secondary *reached* term; the twin’s fit advantage is a **loan called at ≈ 700 steps**.

Per our reporting discipline, results on architecturally-invariant potentials with analytic couplings are labeled **verification**; results on *learned/anharmonic* lattices are labeled **evidence**; results that are properties of the integrator and the autodiff graph rather than of any trained parameter values are labeled **structural**. Every figure caption repeats its grade. All experiments are laptop-CPU (2-dim units, $N \leq 16$ for interference, $N \leq 8$ elsewhere, 3–12 seeds).

2 Setup

The joint map. One dissipative velocity-Verlet step advances the concatenated state $(\mathbf{q}, \mathbf{p})$ by a kick–drift–kick of the joint H then $\mathbf{p} \mapsto (1 - \gamma)\mathbf{p}$. Because H is separable and couplings enter only through positions, the joint step is **conformally symplectic**: $J^\top \Omega J = (1 - \gamma)\Omega$ and $\det J = (1 - \gamma)^d$ on the full $d = \sum_i d_i$ state (per-unit- γ variant $\det J = \prod_i (1 - \gamma_i)^{d_i}$ with the conformality-pairing caveat). Kinetics $T_i(p_i) = \frac{1}{2}p_i^\top M_i^{-1}p_i$ (learned band) or relativistic (imposing $v_i^{\max} = c/\sqrt{M_i}$); coupling $V_c = \kappa_c \|W_i q_i - W_j q_j\|^2$ (learnable W , **static** κ_c) or a small MLP, on chain edges by default.

The relative mode and the price law (theory note, §7.2). A two-unit channel opens a relative mode $\mu_{\text{rel}}^2 = 4\kappa_c/M$; its half-life obeys $n_{1/2} \propto \mu_{\text{rel}}^{-2} \propto \kappa_c^{-1}$ and two units synchronize in $\approx \pi/(2\mu_{\text{rel}} \text{ dt}) \propto \kappa_c^{-1/2}$. This is the **communication price list**: stronger coupling buys faster synchronization at the cost of shorter relative-memory lifetime. Single-unit constants inherited: latch $q_\infty = q_0 + \varepsilon p_0/(M\gamma)$ (infinite half-life, $\gamma > 0$), Noether decay $(1 - \gamma)^n$, crossover $h^* \approx \gamma/2$, mass-independent floor $2 \ln 2/(-\ln(1 - \gamma))$.

Trained lattices. Learned-system results use N -unit chains of 2-dim units, **newtonian_learned** kinetics, **mlp** potentials (hidden 32), spring coupling, $\text{dt} = 0.05$, on **two_timescale_orbits** data whose timescales come from the inertial band alone (shared stiffness $k_i = M_i \omega_i^2 = 1$). Flag-provenance: Appendix A.

3 Results

3.1 The CLU-Net and its guarantees-by-composition

(Designed testbed, analytic couplings $\Rightarrow$ **verification**. Source: *v3-lattice-build* items 1–4, $N \leq 8$, laptop-CPU.)

Per-unit guarantees compose onto the joint state, each a checked contract. **Exact vanishing-coupling reduction**: at $\kappa_c = 0$ the joint step reduces *bit-for-bit* (`jnp.array_equal`, no tolerance) to independent single-unit steps, across heterogeneous dims (2+3), mixed **newtonian_learned/relativistic** kinetics, $\gamma \in \{0, 0.05\}$, empty-edge and $\kappa=0$ -spring graphs. **Joint conformal symplecticity** $< 10^{-12}$ (x64) at $N = 2, 4$, per-unit- γ volume law $\det J = \prod_i (1 - \gamma_i)^{d_i}$ to $< 10^{-12}$. **Communication is charge flow**: the joint Noether charge is conserved to $< 10^{-10}$ at $\gamma = 0$ and decays as $(1 - \gamma)^n$ to $< 10^{-9}$ at $\gamma > 0$, while the single-unit charge is *not* conserved once $\kappa_c > 0$ (drift $> 10^{-3}$). **Mass-anisotropic cap**: $v_i^{\max} = c/\sqrt{M_i}$ to 10^{-9} , heavy band strictly slower.

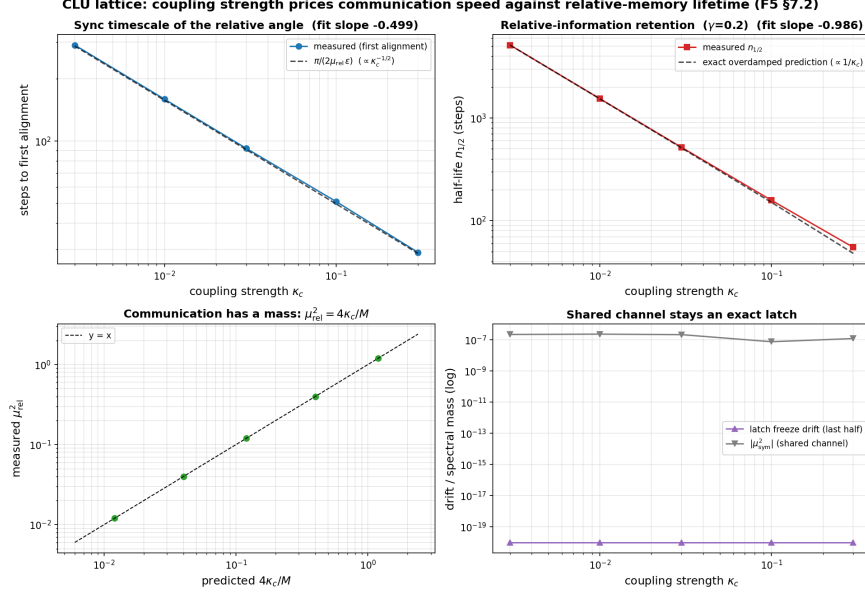


Figure 1: **[verification] The communication price law, on a designed lattice.** Sweeping the static coupling κ_c over two decades on a 2-unit $SO(2)$ pair ($\gamma_{\text{probe}} = 0.2$, $\text{dt} = 0.05$, seed 42): synchronization time follows $\propto \kappa_c^{-1/2}$ (fitted slope -0.499) and relative-memory half-life $\propto \kappa_c^{-1}$ (slope -0.986), matching the theory note’s predicted $-\frac{1}{2}/-1$. Architecturally-invariant potential with analytic coupling $\Rightarrow$ **verification of the theory’s exactness**, not a discovery (v3-lattice-build).

The **communication price law is verified** (designed 2-unit $SO(2)$ pair, $\gamma_{\text{probe}} = 0.2$, seed 42): sweeping κ_c over two decades, $\mu_{\text{rel}}^2 = 4\kappa_c/M$ to six decimals, sync slope -0.499 ($= -\frac{1}{2}$), relative-memory half-life slope -0.986 ($= -1$); the shared channel is a bit-frozen latch (drift 0.0) at every κ_c ; $n_{1/2} = 1544$ at $\mu^2 = 0.04$, $\gamma = 0.2$ reproduces the theory note’s App-N value bit-for-bit. The composition is cheap: joint symplectic error $5.4/4.9/4.4 \times 10^{-8}$ at $N = 2/4/8$ (**constant in N** to within the $f32$ Hessian floor), 233k/125k/35k steps/s. A gated-edge “wormhole” skeleton is present and C^1 (open aligned gate moves a distant force by 3.7×10^{-3} ; closed gate suppresses by five orders, 1.2×10^{-9}) — built, not yet load-bearing (§5).

3.2 The interference firewall, measured: degree-bounded, not width-dependent (headline)

(**Evidence:** lattices with learned MLP potentials. Sources: *v3-interference-extra* (12 seeds, $N \in \{2, 4, 6, 8, 12, 16\}$, scaling + block-monolith controls; 8 seeds, 3 topologies) and *v3-interference-ntk* items 1–3 (3 seeds; κ -sweep, through-training). 2-dim units, MLP potentials, $\kappa_c = 0.05$, measured at initialization unless stated, laptop-CPU. **Headline = Figure 1.**)

The scaling credibility question is catastrophic interference: does updating one unit corrupt the others? We measure it. The operational interference of A on B is the **basin displacement** — the force-field change over B ’s block from a localized CD (wake/sleep) update on A , normalized by the intended change at A :

$$R_{B \leftarrow A} = \frac{\|\Delta F|_B\|}{\|\Delta F|_A\|}, \quad \Delta F = \nabla_q V_{\text{new}} - \nabla_q V_{\text{old}} \text{ after } \theta \leftarrow \theta - \eta[\nabla_\theta V(\text{wake}) - \nabla_\theta V(\text{sleep})].$$

$R_{B \leftarrow A} = 0$ iff B ’s basin is untouched; the aggregate interference *received* per unit is $S_B =$

$\sum_{A \neq B} R_{B \leftarrow A}$ (we report $S = \text{mean}_B S_B$). We compare the **modular lattice** against a **shared-potential monolith** (one $V_\theta : \mathbb{R}^{Nd} \rightarrow \mathbb{R}$ of the same family/width — the honest “no firewall” baseline) and, below, against three *block-diagonal* potentials that separate the two things “modular” could mean.

(i) The structural result: interference is confined to the coupling graph, exactly. Across all 72 modular runs (6 sizes $\times$ 12 seeds), **every non-edge entry of R is exactly zero in float32** — 0 nonzero of 4,656 off-graph entries, against 4,656/4,656 for the monolith. This is a structural identity, not a small number: wake and sleep configurations differ only in unit A ’s block, so $\nabla_\theta[V(q_{\text{wake}}) - V(q_{\text{sleep}})]$ vanishes identically on every own-potential θ_{V_B} ($B \neq A$) and on every coupling module not incident to A ; only θ_{V_A} and the $\leq \deg(A)$ couplings touching A move. Hence

$$S_B = \deg(B) \cdot \bar{R}_{\text{edge}}, \quad \text{verified } S/(\deg \cdot \bar{R}_{\text{edge}}) = 1.000000 \text{ in every topology } \times N \text{ cell.}$$

Per-unit received interference is bounded by the coupling graph’s coordination number, not by the network’s width. A topology control makes this a measurement rather than an inference (8 seeds, $N \in \{4, 8, 16\}$):

topology	degree	$S(N=4)$	$S(N=8)$	$S(N=16)$	slope b (95% CI)	$H_0: b = 0$
chain	$1.50 \rightarrow 1.88$	7.93×10^{-5}	1.50×10^{-4}	1.15×10^{-4}	$+0.264 \pm 0.171$	$p = 0.008$
ring	2.00	1.19×10^{-4}	1.72×10^{-4}	1.32×10^{-4}	$+0.071 \pm 0.183$	$p = 0.391$ (flat)
circulant-4	4.00	—	2.83×10^{-4}	2.64×10^{-4}	-0.066 ± 0.315	$p = 0.636$ (flat)

Non-edge R is exactly zero in *every* topology, including the circulant, where graph distance and index distance disagree: the firewall follows the coupling graph, not the index layout. At fixed degree S is **statistically flat in N** (ring); doubling the degree scales S by $2.01\times$ at $N = 16$ (the identity is exact per-run; the cross-arm ratio inherits the $\pm 30\%$ init scatter of $\bar{R}_{\text{edge}}$, so we quote the identity, never “ $2.0\times$ ” as a constant).

We state the one residual preemptively. A *chain* is not degree-constant: its mean degree ramps $2(N-1)/N = 1.0 \rightarrow 1.88$ over $N = 2 \rightarrow 16$. Its S therefore carries a small real positive slope ($b = +0.46 \pm 0.31$ over $N \in [2, 16]$, 12 seeds; $+0.26 \pm 0.17$, $p = 0.008$, in the 8-seed topology control), which a pure coordination effect predicts as $+0.289$ and which **vanishes under degree normalization** ($b(S/\deg) = +0.17 \pm 0.31$, consistent with 0). We therefore never call a chain “flat in N ”: the claim is *degree-bounded*, and the ring is where flatness is measured.

(ii) The growth result: degree-bounded vs width-linear (Figure 1). With 12 seeds and six sizes the arms’ log-log slopes **diverge**: modular $b = +0.46 \pm 0.31$ vs monolith $b = +1.18 \pm 0.17$ (95% CI, non-overlapping; Welch $p = 3.3 \times 10^{-4}$, paired-by-seed $p = 5.5 \times 10^{-4}$; monolith steeper in 11/12 seeds). The modular curve saturates then decreases ($S(8) = 1.32 \times 10^{-4} \rightarrow S(16) = 1.13 \times 10^{-4}$, $b_{[8,16]} = -0.23 \pm 0.37$); the monolith keeps climbing ($b_{[8,16]} = +0.79 \pm 0.09$).

arm	$N=2$	$N=4$	$N=6$	$N=8$	$N=12$	$N=16$	b over $[2, 16]$
modular (banded $\equiv$ uniform)	6.41e-5	9.76e-5	1.24e-4	1.32e-4	1.12e-4	1.13e-4	$+0.463 \pm 0.311$
monolith (shared V_θ)	2.18e-1	6.40e-1	1.02	1.34	1.90	2.32	$+1.184 \pm 0.170$

(mean $\pm$ s.d. over 12 seeds; separation $\times 20,649$ at $N = 16$.) The monolith’s per-pair displacement is ≈ 0.19 with **no spatial structure**, and its aggregate crosses $S = 1$ at $N \approx 5.9$: **the aggregate**

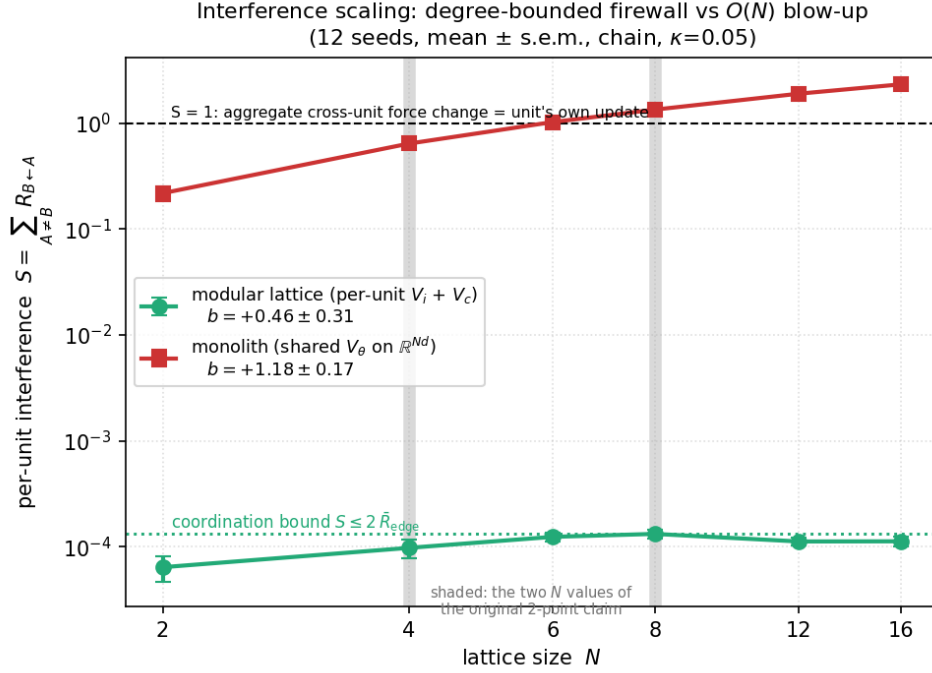


Figure 2: [evidence] **Headline: interference is degree-bounded, not width-dependent.** Per-unit received interference $S = \text{mean}_B \sum_{A \neq B} R_{B \leftarrow A}$ vs lattice size N (log-log; mean $\pm$ s.e.m. over **12 seeds**; chain, $\kappa_c = 0.05$, 2-dim units, MLP potentials, at initialization, laptop-CPU). The shared- V_θ monolith is **width-linear** ($b = +1.18 \pm 0.17$) and crosses $S = 1$ at $N \approx 5.9$ (dashed: aggregate cross-unit force change equals a unit’s own update). The modular lattice tracks the exact identity $S = \text{deg} \cdot \bar{R}_{\text{edge}}$ (its aggregate stays below $2\bar{R}_{\text{edge}}$, dotted, because the chain’s mean degree is < 2) and saturates ($b = +0.46 \pm 0.31$ over $[2, 16]$, entirely the chain’s degree ramp $1.5 \rightarrow 1.88$; $b = -0.23 \pm 0.37$ over $[8, 16]$); slopes diverge, Welch $p = 3.3 \times 10^{-4}$. Shaded: the N range accessible to a two-point measurement, shown to make explicit why two points could not separate these slopes (Appendix G). On the rendered asset’s title, the “ $O(N)$ ” refers to the **monolith**, whose *fitted* exponent is 1.18 ± 0.17 ; no arm of this paper’s modular lattice is described as “ $O(1)$ ” or “flat in N ” in any figure or sentence. Source: `v3-interference-extra`.

cross-unit force perturbation a monolithic unit receives exceeds the magnitude of its own intended update ($S(4) = 0.640$, $S(6) = 1.024$). We phrase this as force perturbation, not storage capacity: reading it as “interferes with itself more than it stores” would require the *dynamical* interference half-life, which we have not measured (Appendix C).

The modular residual leak is coupling-mediated: a κ_c -sweep ($N = 4$, 3 seeds) gives log-log slope 1.99, i.e. $\bar{R}_{\text{edge}} \propto \kappa_c^2$, and **exactly 0 at $\kappa_c = 0$** — a tunable price set by the coupling strength, not a fixed floor. The firewall is **mass-independent** — banded and uniform lattices give *bit-identical* R ($\max |R_{\text{banded}} - R_{\text{uniform}}| = 0.000 \times 10^0$ over 6 sizes $\times$ 12 seeds) — and the separation **persists through training** where measured (2-unit lattices, 3 seeds: modular $R_{\text{off}} 4.2 \times 10^{-5} \rightarrow 8.9 \times 10^{-5}$ over epochs 0/150/300; monolith $O(0.1\text{--}0.4)$ throughout; $N > 2$ through-training unmeasured, Appendix C).

(iii) **What actually buys the firewall: parameter separation, not block structure.** A reviewer’s first question is “is this modularity, or did you just train N nearly-separate networks?” We answer with a control (12 seeds, $N \leq 16$): a **single** potential $V : \mathbb{R}^{Nd} \rightarrow \mathbb{R}$ with an explicitly block-diagonal Hessian, in three parameter-sharing regimes (full table + figures: Appendix H).

arm	params ($N=2 \rightarrow 16$)	$S(N=8)$	$S(N=16)$	per-pair $\bar{R}$	b over $[2, 16]$	non-edge R
block-untied sep. θ_i	2,370 $\rightarrow$ 18,960	0.0	0.0	0.0	undef. ($S \equiv 0$)	0 (exact)
modular lattice ($\kappa_c=0.05$)	2,382 $\rightarrow$ 19,112	1.32e−4	1.13e−4	$\approx 6.5\text{e}−5$	$+0.46 \pm 0.31$	0 (exact)
block-trunk (shared trunk)	1,218 $\rightarrow$ 1,680	4.88e−1	1.05	0.065	$+1.51 \pm 0.36$	nonzero
monolith (cross-block V_θ)	1,253 $\rightarrow$ 2,177	1.34	2.32	0.19	$+1.18 \pm 0.17$	nonzero
block-tied (deep-sets)	1,185 (const.)	6.78	1.45×10^1	0.97	$+1.29 \pm 0.01$	nonzero

Block structure alone recovers nothing: block-trunk and block-tied have strictly block-diagonal Hessians and still show $O(1)$ per-pair interference and width-linear aggregate growth — sharing *any* parameters across blocks, even only a trunk, reopens the channel. **Block structure plus disjoint parameters recovers the firewall completely:** block-untied gives $S \equiv 0$ *exactly*, at every N and every seed (diagonal $R[A, A] = 1.000000$ throughout, so it is a genuine zero, not a dead self-response). **Interference is not a capacity effect:** the worst arm is the 1,185-parameter tied potential (constant in N), the perfect arm the 18,960-parameter untied one — which kills the “you just gave the modular arm more parameters” confound in the cleanest way available.

The honest reading, which we prefer to the flattering one: **nothing physics-specific buys the firewall — parameter separation does, and block-untied is a strictly *better* firewall than our lattice** ($S \equiv 0$ vs $\bar{R}_{\text{edge}} \approx 6.5 \times 10^{-5}$). The modular lattice *is* block-untied plus an $O(\kappa^2)$ graph-local leak, and that leak is exactly the price of having a coupling graph at all (0.0 at $\kappa_c = 0$). What the physics buys is that the lattice pays that price while retaining a **single joint symplectic Hamiltonian with priced, graph-local communication**: units that can talk, under one conservative dynamics, at a measured and bounded cost of talking (§3.3). A network of disconnected potentials has a perfect firewall and nothing to say across it. Our monolith foil is correspondingly *mild*: the permutation-equivariant deep-sets potential $V = \sum_i f(q_i)$ — a real architecture, not a strawman — interferes $6.2\times$ harder in aggregate at $N = 16$, with per-pair $\bar{R} = 0.97$ (an update aimed at A moves every other basin by $\approx 97\%$ of the intended amount, the shared f moving coherently at every block) and exceeds $S = 1$ already by $N = 4$. We report both foils and quote the naive monolith in the headline, which understates our case.

Metric discipline (Appendix D): the raw potential-NTK cosine is ≈ 0.99 for *both* architectures and does not distinguish them; the firewall lives in the wake-sleep difference, which R captures and the raw kernel does not. Report R , never the NTK cosine.

3.3 The price list is predictive, not descriptive

(Evidence: trained 2-unit lattices, prediction registered before measurement. Source: v3-interference-ntk item 2, 3 seeds.)

For each trained lattice we read κ_{eff} from the *learned* coupling potential alone (mass-weighted Hessian of V_c along the relative mode, blind to any rollout) and *then* predict recall horizon and sync from the price law; only after registering do we measure. Varying the static coupling makes κ_{eff} span $91\times$:

κ_{eff} (learned, blind)	sync pred	sync meas	$ \Delta _{\text{sync}/\text{pred}}$	$n_{1/2}$ pred	$n_{1/2}$ meas	$ \Delta _{n_{1/2}/\text{pred}}$
0.0055	210.9	195	7.5%	2773.4	∞ (cens., $> 11,600$)	—
0.0163	123.1	122	0.9%	943.1	1468	55.7%
0.0556	66.6	68	2.1%	273.9	257	6.2%
0.1602	39.2	41	4.5%	92.9	137	47.5%
0.4970	22.3	23	3.2%	27.5	42	52.8%

The **ranking is exact** — $\text{Spearman}(n_{1/2}^{\text{pred}}, n_{1/2}^{\text{meas}}) = 1.0$, $\text{Spearman}(\kappa_{\text{eff}}, n_{1/2}^{\text{meas}}) = -1.0$, both over $n = 5$ lattices (exact $p = 0.008$ under the null; the top-ranked, weakest-coupling lattice’s $n_{1/2}$ is **right-censored** at $> 11,600$ steps and enters as a near-latch, not a point measurement). Sync is predicted **pointwise to $\leq 7.5\%$ relative to the registered prediction** across the $91 \times$ decade (residuals $\{7.5, 0.9, 2.1, 4.5, 3.2\}\%$; the maximum sits on the weakest-coupling lattice, 210.9 predicted vs 195 measured — which is 8.2% if one instead normalizes by the measured value, so we state the denominator). The **recall horizon $n_{1/2}$ itself is predicted in rank, not pointwise**: on the four *uncensored* lattices its residuals are $\{55.7, 6.2, 47.5, 52.8\}\%$ (the first-crossing scatter of a threshold-hitting-time observable), so we claim the *ranking* of $n_{1/2}$ and the *pointwise* value of sync, and nothing stronger. The measured κ_{eff} , read blind from coupling curvature, tells you which trained 2-unit lattice has the longer recall horizon and its sync time before the task is run — demonstrated pointwise at $N = 2$, and, crucially, **not an $N = 2$ artifact: the underlying price law holds flat in N to $N \leq 16$** (Fig. 3, next), so the priced channel spans the same range as the firewall.

The price list is not an $N = 2$ artifact — it holds flat in N to $N \leq 16$ [verification, v3-pricing-n-scaling]. This is the designed- $SO(2)$ /frozen-coupling testbed of Fig. 1 (§3.1) extended across N , so it is labeled *verification of the theory’s exactness across composition*, not a learned-system discovery; the learned-coupling arm is the control below. The underlying price law ($\text{sync} \propto \kappa_{\text{eff}}^{-1/2}$, $n_{1/2} \propto \kappa_{\text{eff}}^{-1}$) holds to its **pre-registered** tolerance not only at $N = 2$ but at $N \in \{2, 4, 8, 16\}$, **both chain and ring, 5 seeds**, on $U(1)$ -preserving (**channel_spring**) trained lattices of designed $SO(2)$ units: fitted sync exponent -0.49 ± 0.02 and $n_{1/2}$ exponent -0.91 ± 0.03 (the ≈ -0.91 vs -1.0 is the pre-registered first-crossing/kick-phase ripple), with μ_{rel}^2 residual $\leq 0.45\%$ and $R^2 \geq 0.998$, and — the load-bearing observation — **flat in N** , with no monotonic drift toward zero and κ_{eff} exact against its analytic value at every N (Fig. 3). The prediction was registered before measurement. This closes the composition gap the firewall/price-list split otherwise opens (§1, §5): the physics-specific priced channel now spans the **same $N \leq 16$ range** at which the degree-bounded firewall is established, rather than living at $N = 2$. Appendix C’s earlier concession that the trained-lattice scaling exponent was “inconclusive” is now **attributed**, not merely retracted: it was an **artifact of the shipped random- W coupling’s $U(1)$ breaking**, which clusters κ_{eff} into a $\leq 1.4\times$ range across a $100\times$ κ -target sweep so no exponent is fittable (control $R^2 \leq 0.47$; App. C), and the symmetry-preserving coupling resolves it — the control’s failure to fit is itself the *explanation* of App. C, not a weakness. *Scope (C-5/C-6)*: designed $SO(2)$ units, a **frozen $U(1)$ -preserving channel_spring coupling** (the one qualifier gating the exponent claim), $N \leq 16$, both topologies, 5 seeds, wake-only training, laptop-CPU.

3.4 Banding as a method, not an oracle gift

(**Evidence:** trained 2-unit ($16\times$ -ratio) lattices, 5 seeds (items 1-2)/3 seeds (item 3, $N \in \{2, 4\}$).
Source: *v3-band-selection 1-3; mass-lr-doctrine-test. Figure 2.*)

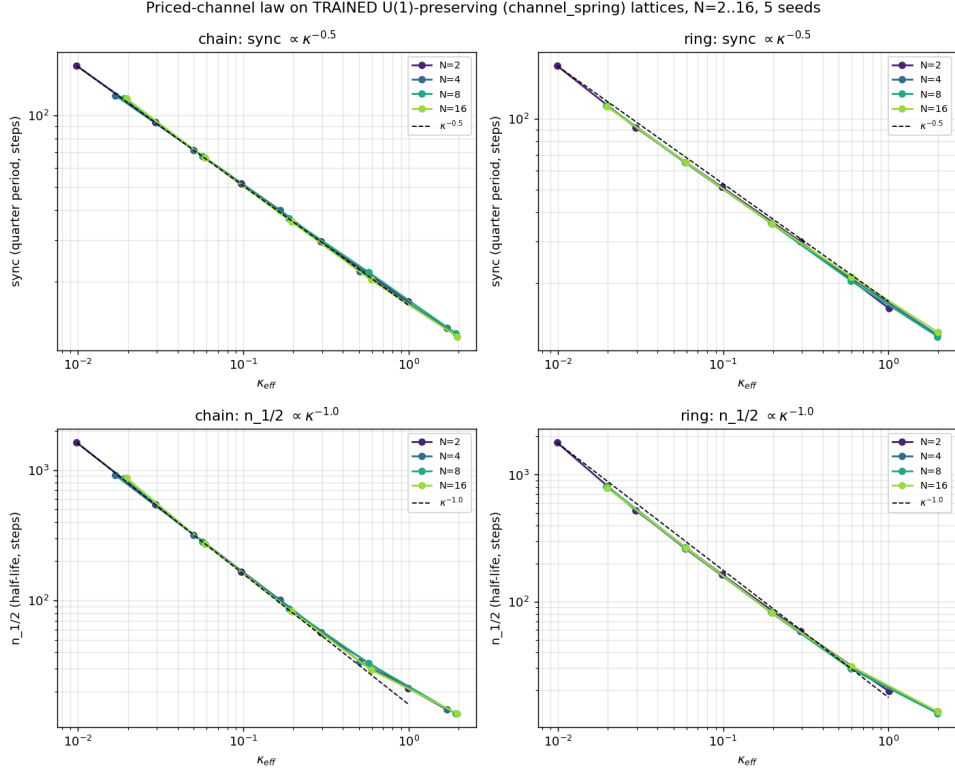


Figure 3: **[verification] The price list holds flat in N to $N \leq 16$.** Per- N log-log fits of synchronization time and recall horizon $n_{1/2}$ against the blind-read κ_{eff} on U(1)-preserving `channel_spring` trained lattices of designed $SO(2)$ units, $N \in \{2, 4, 8, 16\}$, chain and ring, 5 seeds: the price-law slopes (-0.49 ± 0.02 sync, -0.91 ± 0.03 recall) are **flat in N** ($R^2 \geq 0.998$, μ_{rel}^2 residual $\leq 0.45\%$), so the law is not an $N = 2$ artifact. Architecturally-invariant potential with a frozen U(1)-preserving coupling $\Rightarrow$ **verification of the theory’s exactness across composition** (Fig. 1 is the $N = 2$ point of this same testbed). Pre-registered; predictions committed before measurement. Source: `v3-pricing-n-scaling`.

(i) **A wrong band costs.** Four arms spend the *identical* log-spread budget, differing only in placement: matched, anti (inverted), orthogonal (spread on a no-signal axis), uniform. At 300 ep, 5/5 seeds,

$$\text{matched } 1.180 < \text{uniform } 2.416 < \text{orthogonal } 6.924 < \text{anti } 12.791 \quad (\text{held-out MSE}).$$

A correct prior is $0.49 \times$ uniform; an inverted band of identical spread is $5.3 \times$ worse than uniform, $10.8 \times$ worse than matched. Banding is not “any structure helps” and not “the answer for free.”

(ii) **The band is choosable from data — with a stated qualifier.** A per-unit spectral (FFT) estimator maps each dominant frequency to $M_i \propto 1/\omega_i^2$ and recovers the oracle band $[4.0, 0.25]$ with **MSE gap** 0.000 ± 0.000 on 5/5 seeds, one FFT over the training set. *Qualifier:* the zero gap needs well-separated per-unit frequencies; on data whose timescales are not spectrally separable the gap opens — “cheap and exact *when timescales are spectrally separable*.” The optimizer-native alternative (induce-then-snap) is a fragile foil: correct ordering 4/5 seeds, the single inversion pays the full mis-banded penalty (MSE 12.5).

(iii) **The optimizer induces ordering, not magnitude.** Across $N \in \{2, 4\} \times \text{ratio}$

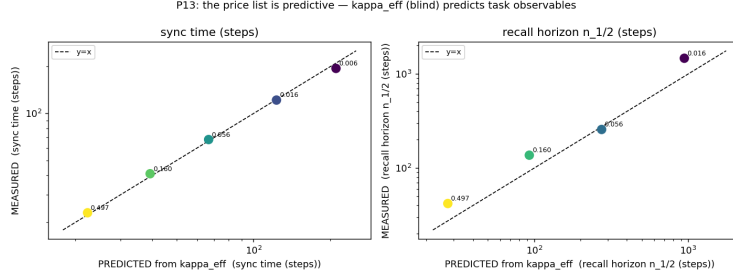


Figure 4: [evidence] **The price list is predictive.** Predicted vs. measured synchronization time across a $91 \times \kappa_{\text{eff}}$ decade on *trained* 2-unit lattices, κ_{eff} read blind from the learned coupling potential and the prediction registered before measurement; points hug the parity line (max residual 7.5% of the prediction; ranking exact, Spearman 1.0). 3 seeds, laptop-CPU (v3-interference-ntk item 2).

$\in \{4\times, 16\times\}$ (3 seeds, 1500 ep), a mass-lr multiplier ≈ 10 reliably induces the correct ordering (alignment $+0.80$ to $+1.00$) and is the safe default; no induced run reaches the designed magnitude (designed fits $7\text{--}14\times$ better). A high multiplier (≥ 30) is harmful in a *ratio- and N -dependent* way: at $16\times$, mult 30 *degrades* the ordering ($+1.00 \rightarrow +0.20 + 0.33$) and $100\times$ *inverts* it (-1.00 at $N = 2$, -0.20 at $N = 4$); at $4\times$ the ordering holds under $100\times$ but a global-mass runaway appears at $N = 4$ (MSE 35.3 ± 47.3 , 3 seeds — s.d. exceeds the mean, a seed-dominated single-run blow-up, not a characterized effect; at $N = 2$, $4\times$, $100\times$ the order holds and MSE is 0.373). Read the hierarchy off alignment/spread, **never MSE** (Appendix E).

3.5 Reversible $O(1)$ -memory training

(Structural measurement on untrained models — a property of the integrator and autodiff graph, independent of parameter values, so not a learned-system claim. Source: v3-reversible-o1 (3 seeds, float32 and float64), CPU, $D \leq 16$, laptop.)

Because the CLU-Net’s separable Hamiltonian gives the dissipative velocity-Verlet step a closed-form exact inverse, and the rollouts are already `lax.scan`-pure, at $\gamma = 0$ training admits **exact recompute-backwards BPTT via analytic leapfrog inversion**: the backward pass reconstructs each previous state by the inverse map instead of storing the forward tape. Reconstructed gradients match standard stored-activation BPTT to $\leq 2.1 \times 10^{-6}$ relative in **float32** (the training default; $\leq 5.8 \times 10^{-15}$ in float64) at T up to 1024 — five-plus orders below typical SGD gradient noise; whether that makes them *training-indistinguishable* is **untested**, since these gradients have not yet been used to train (§5). Peak activation memory is $O(1)$ **in sequence length vs $O(T)$** : at $T = 1024, N = 2$ the reversible pass uses 6.3 KB against the standard 6.00 MB — a $946\times$ **reduction** — the standard cost growing linearly in T while the reversible cost stays flat (scaling only with state size). Wall-time is $\approx 0.9\times$ standard (parity), but **only in the CPU / small- D regime measured here** ($D \leq 16$, small potential MLP, laptop); on a GPU or with large potentials the recompute would likely surface as up to $\sim 2\times$ overhead, so we claim parity only for this regime while the $O(T) \rightarrow O(1)$ *memory* scaling is structural and does generalize.

Exact only at $\gamma = 0$. Any $\gamma > 0$ makes the $(1 - \gamma)$ contraction invertible only in exact arithmetic and amplifies reconstruction error as $(1 - \gamma)^{-n}$, giving a *finite* usable reversal horizon: $\approx 3.3 \times 10^4$ steps at $\gamma = 10^{-3}$ and $\approx 5 \times 10^2$ at $\gamma = 0.05$ (float64; $\sim 100\times$ shorter in float32). This dovetails with the budget framing: the **reversibly- $O(1)$ -trainable memory is exactly the conservative**

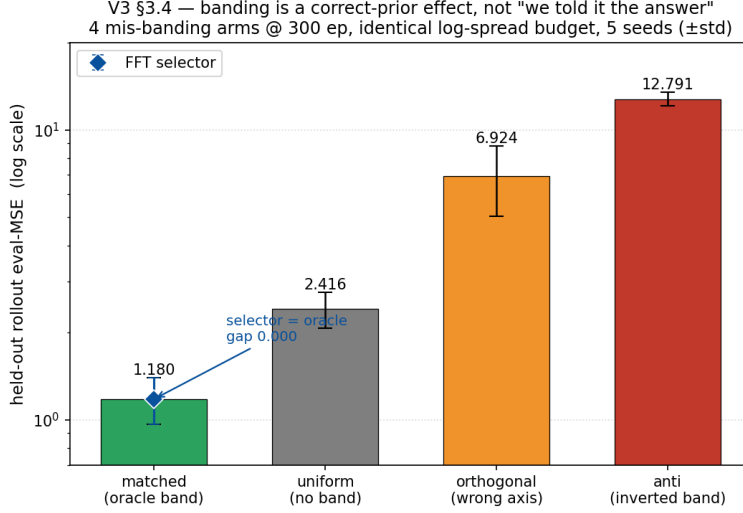


Figure 5: [evidence] **Banding is a correct-prior effect, not an oracle gift.** Held-out rollout MSE at 300 epochs (log scale, 5 seeds, $\pm$ s.d.) for four arms spending the *identical* log-scale-spread budget on a 2-unit lattice at $16\times$ ratio: matched (oracle band) $1.180 \pm 0.216 <$ uniform (no band) $2.416 \pm 0.345 <$ orthogonal (spread on a no-signal axis) $6.924 \pm 1.883 <$ anti (inverted band) 12.791 ± 0.698 ; matched beats every arm on 5/5 seeds. Diamond: the from-data FFT selector lands on the oracle band (selector-oracle MSE gap 0.000 ± 0.000 , 5/5 seeds) — *when timescales are spectrally separable* (v3-band-selection items 1–2).

($\gamma = 0$) **memory** — *reversibly-trainable memory is conservative memory* — while dissipative segments fall back to stored activations or checkpointing (§4). *Scope (C-6)*: this is the gradient mechanics only, on untrained models with a final-state loss; it is **not yet wired into the shipped train_chlu** (windowed per-step loss), so a structural systems result, not yet a trained-lattice measurement (§5).

3.6 The price of the physics prior, measured through the horizon

(*Evidence: learned MLP-potential controls, param-matched, on a single CLU unit ($N = 1$) — the price of the per-unit prior, not a lattice measurement. Source: *minus-the-physics Part A; fit-gap-anatomy* 1–3. Single unit ($N = 1$), dim 2–4, synthetic, laptop-CPU.)*

Symplectic volume conservation buys **bounded (BIBO) dynamics, a protected near-flat memory direction, and vacuum robustness under CD training**; the leapfrog structure buys the **latch**; the physics prior **costs raw fit**. Against a param-matched ($\pm 0.05\%$) unconstrained twin and a broken-volume ablation: BIBO settled-fraction 1.00 (CLU) / 0.33 (broken-vol) / 1.00 (twin); protected flat μ^2 0.008 / 0.122; latch coset drift 0.19 (CLU) vs 1.15 (twin, $\sim 6\times$ looser); the vacuum *survives* CD for the CLU where the broken-volume arm collapses to $r^* = 0$. *Two entries need care.* The BIBO claim is against the broken-volume ablation (1.00 vs 0.33): the twin *matches* the CLU’s settle-fraction (1.00) on this short settle-probe but is bounded only empirically and *diverges* on long rollout (MSE 196 at 5000 steps, Appendix F.1), whereas the CLU is bounded by construction. The cost is specifically **contraction-forbidden** (volume; the broken-vol arm recovers $\approx 2.4\times$ fit by *leaking* volume), **not** the causal cap (inactive here): the twin fits $\approx 15\times$ better (0.013 vs 0.190). (*These are the **minus-the-physics** dim-4 numbers; the recovery ladder in Appendix F.2 uses a distinct circle-vacuum probe — twin 0.0047, CLU 0.207, latch drift 0.778 — on a different testbed,*

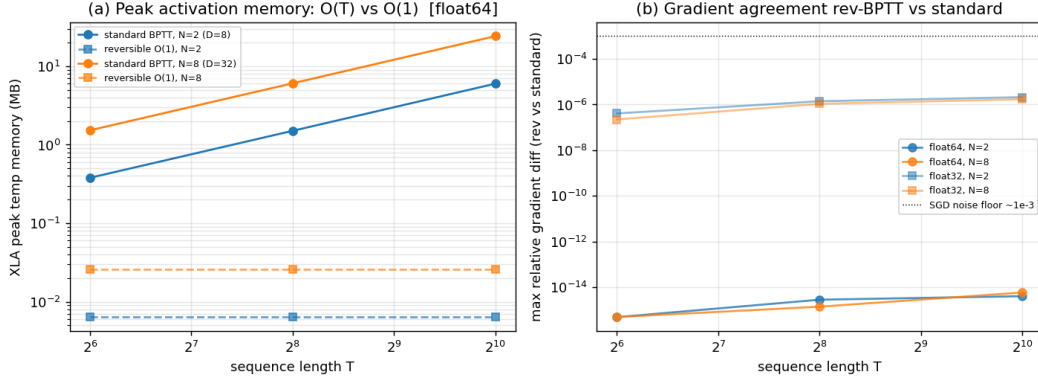


Figure 6: **[structural] Reversible $O(1)$ -memory training ($\gamma = 0$).** (left) peak activation memory vs. sequence length T : standard BPTT grows $O(T)$ while recompute-backwards stays flat $O(1)$ -in- T ($946\times$ at $T=1024, N=2$); (right) reconstructed-gradient relative difference vs. T in float32/float64 ($\leq 2.1 \times 10^{-6}$ f32, $\leq 5.8 \times 10^{-15}$ f64 — five-plus orders below typical SGD gradient noise; training-indistinguishability untested, §5). **Structural measurement on untrained models** (a property of the integrator and the autodiff graph), CPU, $D \leq 16$, 3 seeds; exact only at $\gamma = 0$; the memory metric is the XLA compiler scratch estimate, a proxy for tape size rather than runtime peak RSS (Appendix A.6). (v3-reversible-o1.)

not comparable digit-for-digit to the dim-4 pair here.)

The loan is called at ≈ 700 steps. Against the matched control on a stationary vacuum, the twin leads by 1–2 orders out to ≈ 500 steps, is overtaken at ≈ 700 , then diverges (cumulative MSE 196 at 5000 steps) while the CLU is the only arm flat *by construction* (bounded plateau 0.20–0.23). We claim the crossing and boundedness-by-construction; we do *not* claim the lowest long-horizon plateau (broken-vol 0.14, LSTM 0.13 sit below CLU 0.22). The **reach** term is real but secondary and aggregate-only: with the cap active it costs +77% MSE at $c = 0.5$ and collapses to the uncapped baseline once $v_{\max} \geq$ required speed ($c \geq 2$), with no per-step far-region concentration ($\text{corr}(\text{err}, |\dot{q}|) \approx 0$).

Paid mechanisms rent back specific cheats (not interchangeable). On the circle-vacuum probe (Appendix F.2), a **global γ** recovers 92% of the **absolute contraction-forbidden fit gap, with BIBO, latch, and μ^2 all preserved** ($\det J = (1 - \gamma)^d$); the recovered unit still **trails the twin by $\approx 4.6\times$ in ratio** (0.0216 vs 0.0047 eval-MSE), but is bounded by construction (V2’s exact wording for this shared number). A position-gated field γ_φ does *not* recover fit (-24%) — the wrong lever. A squeeze $S^{(M)}$ rents reach with an exact **matched-quadratic- H certificate** (oracle placement; $\det = 1$ to 3×10^{-7} , injection $\leq e^{2|\zeta|}$, $\zeta = 1$ grants $\Delta q = 2.18 > 2.0$). Ladder + negatives: Appendix F.

4 Related work

Hamiltonian and symplectic recurrence (home turf). Our unit is a *learned-Hamiltonian, symplectically-integrated* recurrent cell, so the nearest architectural cousin is the **Symplectic Recurrent Neural Network** (Chen, Zhang, Arjovsky & Bottou, 2020), which learns a Hamiltonian

and advances it with a symplectic integrator under multi-step training; Hamiltonian and Lagrangian Neural Networks (Greydanus, Dzamba & Yosinski, 2019; Cranmer et al., 2020) learn the energy from data, while stability-oriented recurrences — Lipschitz RNNs (Erichson et al., 2021) and the coupled-oscillator coRNN (Rusch & Mishra, 2021) / LEM (Rusch et al., 2022) — engineer bounded dynamics without a symplectic form. The classic gated baseline our Appendix F.1 compares against is the LSTM (Hochreiter & Schmidhuber, 1997). Against all of these our contribution is deliberately orthogonal: **not a better single cell but a *composition law* for many such cells** — the measured interference firewall (§3.2), the coupling price list (§3.3), and the reversible- $O(1)$ corollary (§3.5) are statements about *coupling* learned-Hamiltonian units into a lattice, a regime none of the single-cell works address. The conformal-symplectic (damped-leapfrog) structure and its $h < 2$ stability limit are standard for dissipative geometric integrators (Hairer, Lubich & Wanner), read here as *memory* statements for a learnable H . **Symmetry and memory lifetime.** Mo (2026) proves an exactly equivariant recurrent *flow* has $\dim(G/H)$ zero Lyapunov exponents and that a “pseudo-gap” predicts a finite lifetime — the *kinematics* of protection, silent on what *sets* the gap; our lattice supplies a constitutive, mass-controlled gap and a measured account of how protection composes across many coupled units. **Geometry-first structured recurrence.** A line engineers recurrence from a chosen symmetry (Di Bernardo et al.; Keller); we headline the *pricing/budget* consequences of a coupling, not “we choose G/H .” **Modular vs. monolithic sharing.** Catastrophic interference — overwriting stored function when a distributed network learns new content — has been understood since McCloskey & Cohen (1989) and French (1999) as a consequence of *shared* representations. Two remedy families exist. The first *prevents* interference by construction: modular architectures and mixtures-of-experts route content to different parameters (Jacobs et al., 1991; Shazeer et al., 2017), and parameter-isolation continual learners freeze or mask per-task subnetworks (Kirkpatrick et al., 2017; Mallya & Lazebnik, 2018). The second *measures* interference to minimize it during optimization: the NTK-overlap matrix quantifies cross-task forgetting (Doan et al., 2021), and gradient-alignment/gradient-surgery methods treat conflicting (negative-inner-product) updates as the signal to suppress (Riemer et al., 2019; Yu et al., 2020). Our lattice sits at the intersection: interference between units is neither merely prevented nor globally minimized but *measured as a coupling-resolved kernel*. A shared-potential monolith moves every other unit’s basin by ≈ 0.19 per update, unstructured and **width-linear** ($N^{1.18 \pm 0.17}$); the modular lattice moves a graph neighbour by $\approx 6.5 \times 10^{-5}$ and a non-adjacent unit by *exactly* zero, the leak $\propto \kappa^2$ (zero at $\kappa = 0$) — an architectural **degree-bounded vs. width-linear** separation and explicit coupling power-law, a *priced*, graph-local firewall rather than the diffuse, data-dependent overlap on monoliths (Doan et al., 2021) or the sample-complexity separation for modular tasks (Boopathy et al., 2025). Our controls locate the mechanism precisely, and not where a physics-first reading would put it: **parameter separation, not block structure or geometry, closes the channel** — a block-diagonal potential with tied parameters (a permutation-equivariant, deep-sets-style summed potential) interferes *worse* than the unstructured monolith, while the same potential with per-block parameters gives exactly zero interference. Our firewall is thus the parameter-isolation family’s guarantee (Kirkpatrick et al., 2017; Mallya & Lazebnik, 2018) obtained *by construction* rather than by penalty or mask; what the joint Hamiltonian adds is not a better firewall but a **priced channel through it** (§3.3), so that units may communicate under a single conservative dynamics at a coupling cost that is measured, graph-local and $O(\kappa^2)$. (*Scope: 2-dim units, chain/ring/circulant-4, MLP potentials, $N \leq 16$, $\kappa = 0.05$, 8–12 seeds, measured at initialization; §3.2, Appendices D and H.*) **Reversible / $O(1)$ -memory training.** Trading stored activations for recomputation is the standard route to memory-cheap backprop: gradient checkpointing recovers activations at $O(\sqrt{T})$ memory, and reversible architectures (RevNet / momentum-net / reversible-integrator lineage) reconstruct them exactly at $O(1)$ by an invertible forward map. The damped-symplectic lattice is invertible for the

physics reason that a separable Hamiltonian’s leapfrog step has a closed-form inverse; §3.5 *measures* the resulting trade (exact $O(1)$ -in- T memory at $\gamma = 0$, float-indistinguishable gradients) and makes explicit the boundary the general framing hides: exactness holds only along the conservative direction, so a network interleaving $\gamma > 0$ blocks must checkpoint those segments. Positioning against the checkpointing $O(\sqrt{T})$ middle-ground and accelerator-scale measurement are open (§5). (Specific published anchors — checkpointing, RevNet/momentum-net — to be finalized from the scout bibliography.)

5 Limitations and horizon

Scope. Every learned-system number is on 2-dim units, MLP potentials, laptop-CPU, synthetic two-timescale data: $N \leq 16$ and 8–12 seeds for interference, $N \leq 8$ and 3–5 seeds elsewhere; industrial-scale and real-data behavior are untested (the largest gap). **The composition, owned (so no reviewer has to assemble it).** Two facts sit far apart in this paper and a reader will compose them, so we do it here. (a) The interference firewall is our headline but **not our physics**: a potential of disjoint per-unit parameter blocks is a strictly better firewall than our lattice ($S \equiv 0$ vs $\bar{R}_{\text{edge}} \approx 6.5 \times 10^{-5}$, §3.2(iii)) with no Hamiltonian at all — *parameter isolation*, not the symplectic structure, confines interference. (b) Our physics-specific claim is the **priced channel** (§3.3), and it is **no longer stranded at $N = 2$** : its predictive parity is demonstrated on trained 2-unit lattices with *learned* couplings (evidence, §3.3), and the underlying price law is now **verified to hold flat in N at $N \in \{2, 4, 8, 16\}$** , both topologies, on the designed- $SO(2)$ /frozen- $U(1)$ -preserving-coupling testbed — the same substrate as the $N = 2$ verification (§3.1), extended across composition (Figs. 3–7; pre-registered, 5 seeds). Appendix C’s earlier “inconclusive on trained lattices” is resolved: it was an **artifact of a symmetry-breaking coupling** (a *learned* random- W coupling clusters κ_{eff} into a $\leq 1.4\times$ range so no law is fittable), which the $U(1)$ -preserving coupling fixes — the control’s failure is the *explanation*, not a weakness. Composed: the degree-bounded firewall is established (evidence) at $N \leq 16$, and the priced channel it lets through is now verified across the **same** $N \leq 16$ range, so the composition gap is closed — the physics-specific contribution spans the firewall’s range rather than living at $N = 2$. The residual honest scope: the exponent’s cross- N authority is the designed testbed (as at $N = 2$), and reading the *fitted exponent* off *freely-learned* couplings at $N > 2$ (rather than the ranking/parity map, which already survives learning) remains a follow-up — but no main-text claim now bounds on an $N = 2$ limitation. The thesis composes without a residual: **modularity is the firewall; a single joint symplectic Hamiltonian is what lets the modules talk, at a price we measure — across the full range at which the firewall is established. Certificate altitude.** The “exactly zero” beyond the graph is a property of the position-only coupling and the locality of the contrastive update (it holds on chain, ring and circulant-4 graphs, and is analytic, not fitted); it is measured on the potential NTK (basin displacement ΔF), not a full attractor re-settle, so the $S = 1$ crossing is force perturbation and not storage, and the *dynamical interference half-life* is **unmeasured** (Appendix C). Interference is measured **at initialization**: the exact zeros are training-invariant by construction, but $\bar{R}_{\text{edge}} \approx 6.5 \times 10^{-5}$ is an init-scale quantity and through-training persistence is measured only at $N = 2$. A *trajectory-mediated* interference channel is **unmeasured**. The arms are not parameter-matched (19,112 modular vs 2,177 monolith at $N = 16$); the confound is bounded rather than eliminated by the block-tied (1,185 params, worst) / block-untied (18,960 params, zero) controls. The $\propto \kappa^2$ leak law is established at $N = 4$ only. The pricing prediction is on 2-unit trained lattices; the selector’s zero gap is contingent on spectral separability; the wormhole edge is built and C^1 but not shown to beat a dense-MLP baseline, and the intra-unit

wormhole is a design hypothesis. **Reversible-training scope.** The $O(1)$ -memory result (§3.5) is the gradient mechanics on *untrained* models with a final-state loss on CPU; it is **not yet wired into train_chlu**, the wall-time parity is CPU/small- D only, and exactness is $\gamma = 0$ -only — wiring it into the real trainer, an accelerator (GPU/HBM) memory + honest wall-time measurement at $T \gtrsim 4k$ where standard BPTT OOMs, and a gradient-checkpointing ($O(\sqrt{T})$) Pareto baseline are the load-bearing follow-ups. **Horizon:** those reversible follow-ups; through-training interference at $N \in \{4, 8, 16\}$ and a parameter-matched monolith; a κ -sweep at $N = 16$ to close $S = \deg \cdot c \kappa^2$ into a closed-form price for the firewall; 2-D grid and small-world topologies (where a wormhole edge should add exactly its own degree contribution — a *quantitative price for a wormhole*); the irrep firewall; real-data banding.

A Flag-provenance tables (C-7)

All results laptop-CPU, JAX; numbers transcribed from source reports. **A.1 CLU-Net (§3.1, v3-lattice-build, verification):** commits c124103/9c39b12/79dcac9; chain, SpringCoupling (static κ_c); pricing run designed $SO(2)$ pair $\gamma_{\text{probe}} = 0.2$, seed 42, f_{32} , $\kappa_c \in \{0.003, 0.01, 0.03, 0.1, 0.3\}$; x_{64} tests; scaling smoke $N \in \{2, 4, 8\}$. **A.2 Firewall, κ -sweep and through-training (§3.2, v3-interference-ntk):** 9a13455, chlu 0.2.4, seeds $\{0, 1, 2\}$; unit_dims $[2] \times N$, mlp hidden 32, newtonian_learned; monolith CHLU(dim=2N); probe $\eta = 0.05$, $r = 0.5$; $\kappa = 0.05$, sweep $\{0, 0.01, 0.03, 0.1, 0.3\}$ at $N = 4$; through-training epochs $\{0, 150, 300\}$ at $N = 2$. *Supersession:* this run’s $N \in \{4, 8\}$, 3-seed scaling numbers are superseded for the *growth* claim by A.2b (12 seeds, 6 sizes); the two harnesses agree bit-exactly on the shared cells (max rel diff 0.00×10^0), so differences are seed sampling, not metric drift (Appendix D.1). **A.2b Interference scaling, topology and block-monolith controls (§3.2(i)–(iii), v3-interference-extra, evidence):** commit 37dc664, repo read-only (git status --porcelain empty before & after); main venv, **JAX 0.9.0**, CPU (CpuDevice(id=0)), equinox, chlu 0.2.4, scipy 1.17.0, x32 default precision; seeds $\{0 \dots 11\}$ (12) for scaling/block, $\{0 \dots 7\}$ (8) for topology; N grid $\{2, 4, 6, 8, 12, 16\}$ (scaling/block), $\{4, 8, 16\}$ (topology); build_lattice, unit_dims $[2] \times N$, mlp hidden 32, newtonian_learned, spring coupling, coupling_dim 2, proj_init_scale 0.1, $\kappa_c = 0.05$, edges chain (default)/ring/circulant-4; banding mass_scales = 4.0 (even) / 0.25 (odd), uniform = None; monolith CHLU(dim=2N,hidden=32); block monoliths: single $V : \mathbb{R}^{2N} \rightarrow \mathbb{R}$, per-block MLP $\text{Linear}(2,32) \rightarrow \tanh \rightarrow \text{Linear}(32,32) \rightarrow \tanh \rightarrow \text{Linear}(32,1) + 0.05\|q\|^2$ (mirrors PotentialMLP), regimes block_untied/block_trunk/block_tied; probe = *identical code path* to A.2 (force_interference imported verbatim), $\eta = 0.05$, $r_{\text{probe}} = 0.5$, all units at memory loci, only A perturbed (wake = m_A , sleep = $m_A + 0.6N$), R row-normalized by $R[A, A]$; measurement at **initialization** (no training in this run); NTK cosine not computed; non-defaults kinetic=newtonian_learned (banding needs log_mass), training config inapplicable; *metric-identity anchor:* re-running this harness on seeds $\{0, 1, 2\}$ at $N \in \{4, 8\}$ reproduces v3-interference-ntk/interference_init.json to max relative difference 0.00×10^0 (bit-exact) for both arms; statistics = per-seed OLS of $\log_{10} S$ on $\log_{10} N$, mean $\pm 95\%$ CI over seeds (t , dof 11), Welch and paired-by-seed t -tests for the slope difference. **A.3 Pricing (§3.3):** wake-only, lr 10^{-3} , x_{64} , $\kappa_{\text{static}} \in \{0.01, 0.03, 0.1, 0.3, 1.0\}$, 200 ep. **A.3b Price law flat in N (§3.3, Figs. 3–7, v3-pricing-n-scaling, verification across composition, pre-registered):** commit e3c8931 (main), JAX 0.9.0, CPU, x_{64} ; wake-only (sleep off), $M = 1$, dt = 0.05, 60 ep, window 128, $\gamma = 0$ (sync)/ $\gamma_{\text{probe}} = 0.2$ ($n_{1/2}$); designed $SO(2)$ MexicanHat units ($f=M=\lambda=1$, start at analytic vacuum, $\kappa_{\text{eff}} 0.1981 \rightarrow 0.1983$ over 60 ep); *primary channel_spring* ($U(1)$ -preserving, frozen), $N \in \{2, 4, 8, 16\} \times \{\text{chain, ring}\} \times \kappa_{\text{target}} \in \{0.01, 0.03, 0.1, 0.3, 1.0\} \times \text{seeds } \{0-4\}$ (198/200 rows;

ring/ $N=16/\kappa=1.0$ is 3 seeds, ± 0.014); *control* **spring_random** (legacy random- W , $U(1)$ -breaking, trainable W , static $\kappa_S = 0.1$), chain, 100 rows; extractor $\kappa_{\text{eff}} = \frac{1}{4}M\lambda_{\text{max}} = 0.5\kappa\lambda_{\text{max}}(L)$ (5-dp identity); pre-registered decision rule (HOLDS iff $s_N \in [-0.55, -0.45]$, $h_N \in [-1.15, -0.80] \forall N$, no drift, $\mu_{\text{rel}}^2 < 2\%$) applied verbatim; non-defaults **coupling_type=channel_spring**, langevin n/a (wake-only). **A.4 Banding (§3.4)**: band-selection 9a13455, module c4bc004 (additive), **mass_lr_mult** $\in \{1, 3, 10, 30, 100\}$, seeds items 1/2 {0..4}, item 3 {0,1,2}; *env caveat*: band-selection work-tree JAX 0.10.2 vs main 0.9.0 (headline numbers version-stable, reproduce b1782b0 bit-for-bit); mass-lr db3369b, commits be12b73/df11483, data masses [4.0, 0.25]. **A.5 Price of physics (§3.6)**: minus-physics db3369b (9533ef3/c33b41f/b41410f), dim 4, hidden 64, mlp, 150 ep, seeds {42, 43, 44}, wake-only, sleep_friction 0; fit-gap 9a13455; param match chlu 4549 / broken_volume 4557 / twin 4551. **A.6 Reversible $O(1)$ (§3.5, v3-reversible-o1, structural)**: commit 63fea62 (main @ wave-6), JAX **0.9.0**, device cpu, repo read-only (inverse map reconstructed from H , validated against the shipped **velocity_verlet_step**); item 1 **CHLU(dim=8,hidden=16)**, **newtonian_learned**, mlp, *untrained* random init, $\gamma \in \{0, 10^{-4}, 10^{-3}, 10^{-2}, 0.05, 0.1, 0.3\}$, float32 and float64, T to 16384; item 2 **build_lattice([2] $\cdot N$)** $N \in \{2, 8\}$ hidden 16, spring $\kappa_c = 0.05$, chain, seeds {0, 1, 2}, final-state MSE, $T \in \{64, 256, 1024\}$, *untrained*; memory via XLA **memory_analysis().temp_size_in_bytes** (compiler estimate, deterministic; proxy for tape size); grad rel-diff $\leq 2.1 \times 10^{-6}$ (f32)/ $\leq 5.8 \times 10^{-15}$ (f64) @ $T=1024$; peak temp 6.00 MB vs 6.34 KB (946 $\times$, $N=2$), 24.20 MB vs 25.7 KB (940 $\times$, $N=8$); wall-time 0.86 $\times$ (f64)/0.94 $\times$ (f32); $\gamma > 0$ f64 horizons $\geq 1.3e5/3.3e4/4.1e3/5.1e2/1.3e2$ at $\gamma = 10^{-4}/10^{-3}/10^{-2}/0.05/0.1$ ($\sim 100\times$ shorter f32). *Scope*: untrained + final-state loss (not the shipped windowed trainer), CPU $D \leq 16$; wall-time regime-specific; memory scaling structural.

B The mis-banding price panel and over-lr failure modes (C-9)

B.1 Degradation curve (5 seeds, v3-band-selection item 1):

budget	matched	uniform	orthogonal	anti
60 ep	2.812 ± 0.327	3.751 ± 0.336	8.874 ± 2.198	14.905 ± 0.686
300 ep	1.180 ± 0.216	2.416 ± 0.345	6.924 ± 1.883	12.791 ± 0.698

matched beats every arm 5/5 seeds at both budgets (300-ep ratios 0.488/0.170/0.092). **B.2 Selection recipe (item 2)**: uniform 2.416 ± 0.345 ; oracle 1.180 ± 0.216 ; **selector** 1.180 ± 0.216 (**gap** $+0.000 \pm 0.000$); masslr-init 3.527 ± 4.493 (ordering 4/5; seed-2 inversion $\rightarrow$ MSE 12.5). **B.3 Over-lr (N8/N9)**: at 16 $\times$, 100 $\times$ lr $\rightarrow$ alignment -1.00 , unitM (3.90, 5.34), *lowest* MSE 0.324 (a trap — N8); at 4 $\times$, 100 $\times$ lr $\rightarrow$ global runaway MSE 35.3 ± 47.3 ; slow-first curriculum did not help (align -0.54 ± 0.72 ; N9). Safe point: multiplier ≈ 10 .

C Honest unmeasured list (open items)

Trajectory-mediated banding channel (banding changes which loci drive updates — unmeasured); *irrep firewall* (does the NTK factor across irreps? — unmeasured); *block-structured monolith* — **now MEASURED** (§3.2(iii), Appendix H): it is the separate *parameters*, retained here as a record of the item’s discharge; *dynamical interference half-life* (we map ΔF , not the event’s dynamical half-life — unmeasured; the one thing still blocking a *storage* reading of the $S = 1$ crossing); *through-training interference at $N > 2$* (R measured at init for $N \leq 16$; exact zeros training-invariant by construction, but $\bar{R}_{\text{edge}}$ is init-scale and only $N = 2$ was tracked to 300 epochs —

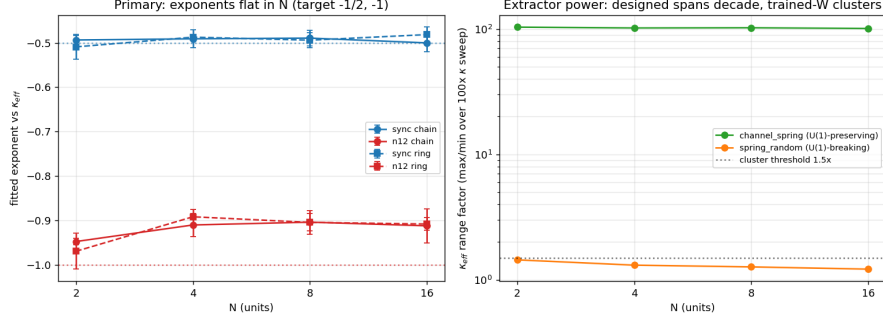


Figure 7: [verification/control] The price-law exponents are flat in N ; the $U(1)$ -breaking control is unfittable. Fitted sync and $n_{1/2}$ exponents vs lattice size $N \in \{2, 4, 8, 16\}$ on the $U(1)$ -preserving `channel_spring` arm (chain and ring, 5 seeds): both cluster on the theory’s $-\frac{1}{2}/-1$ with no monotonic drift (sync -0.49 ± 0.02 , $n_{1/2}$ -0.91 ± 0.03 , $R^2 \geq 0.998$). The `spring_random` control ($U(1)$ -breaking) is off-scale with 95% CIs spanning ± 12 to ± 887 — the direct cause of the former App. C “inconclusive,” now resolved. Pre-registered decision rule applied verbatim. Source: `v3-pricing-n-scaling`.

unmeasured at $N \in \{4, 8, 16\}$); *parameter-matched monolith* (19,112 vs 2,177 params at $N = 16$; capacity is decoupled from interference by the tied/untied pair, but a width-matched sweep is unrun); κ -sweep beyond $N = 4$ (would close $S = \deg \cdot c \kappa^2$ into a closed-form price — unmeasured); *non-MLP potentials and richer graphs* (no `so2_invariant/deep_mlp`; no 2-D grid or small-world topologies — unmeasured); *trained-coupling scaling exponent* — **now RESOLVED** (N16; §3.3, Figs. 3–7, `v3-pricing-n-scaling`): the earlier “inconclusive” is attributed to the shipped random- W coupling’s $U(1)$ breaking (clusters κ_{eff} to a $\leq 1.4\times$ range, control $R^2 \leq 0.47$, exponents ± 12 to ± 887), *not* a physics degradation; on a $U(1)$ -preserving `channel_spring` coupling (designed $SO(2)$ units, frozen coupling) the exponent is **verified flat across** $N \leq 16$ (-0.49 ± 0.02 / -0.91 ± 0.03 , both topologies, 5 seeds, pre-registered) — the N -extension of the §3.1 verification — so the “inconclusive” is understood as a property of the *learned* symmetry-breaking coupling, not of N (reading the fitted exponent off freely-learned couplings at $N > 2$ remains open; the ranking/parity map already survives, §3.3); retained as a record of the item’s discharge; *intra-unit wormhole* (design hypothesis; discriminating test vs dense-MLP not yet run).

D Metric discipline: why not the NTK cosine (CM-9 tail); and the resolved-spatial-structure panel

The potential-NTK cosine $\cos \angle(\nabla_{\theta} V(q_A), \nabla_{\theta} V(q_B)) \approx 0.99$ for *both* architectures (shared low-level features), hence uninformative. The firewall lives in the wake–sleep difference, where off-unit gradients cancel for the modular lattice and not for the monolith; $R_{B \leftarrow A}$ captures exactly this. Reviewers must use R , not the raw kernel.

D.1 Spatial structure of R (3 seeds, $N \in \{4, 8\}$; `v3-interference-ntk` items 1/3). Retained because it is the only place the *spatial* structure (neighbour vs non-adjacent) is printed per-arm; its aggregate S values are superseded, for the growth claim, by the 12-seed six-size measurement of §3.2(ii).

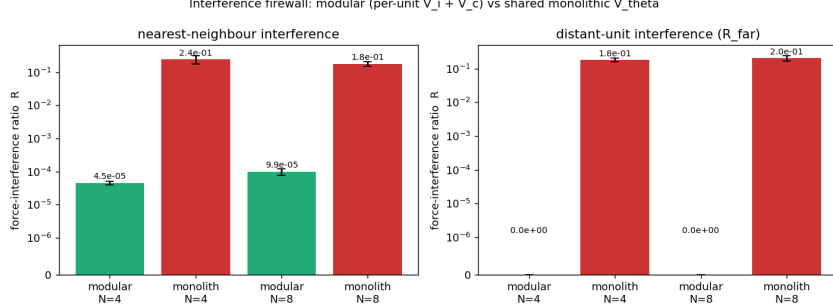


Figure 8: [evidence] **Spatial structure of the interference kernel.** Cross-unit basin displacement R resolved by graph distance for the modular lattice ($\approx 10^{-5}$ neighbour, exactly zero non-adjacent) vs. the shared-potential monolith (≈ 0.2 , spatially unstructured: $R_{nn} \approx R_{far}$), $N \in \{4, 8\}$, 3 seeds, laptop-CPU (v3-interference-ntk items 1/3; aggregate growth superseded by Fig. 2).

config	R_{off} mean	R_{nn} (nbr)	R_{far} (non-adj)
modular $N = 4$	2.26×10^{-5}	4.53×10^{-5}	0.000 (exact)
monolith $N = 4$	2.12×10^{-1}	2.42×10^{-1}	1.82×10^{-1}
modular $N = 8$	2.48×10^{-5}	9.92×10^{-5}	0.000 (exact)
monolith $N = 8$	1.98×10^{-1}	1.78×10^{-1}	2.04×10^{-1}

Reconciliation (no contradiction; C-7). These are 3-seed values ($\{0, 1, 2\}$); §3.2 quotes 12-seed values from a harness that reproduces this JSON **bit-exactly on the shared seeds and sizes** (max rel diff 0.00×10^0). The differences (per-edge leak 4.53×10^{-5} here vs $\bar{R}_{edge} = 6.51 \times 10^{-5}$ at $N = 4$ over 12 seeds) are seed sampling of an init-scale quantity with $\pm 30\%$ scatter, not metric drift. **A negative worth recording (C-9):** on 3 seeds the $N = 4 \rightarrow 8$ *growth factors* were modular $\times 2.56$ vs monolith $\times 2.18$ — an apparent “same slope, different offset” reading. The per-seed 95% CI on that modular growth factor is ± 2.27 : the comparison was pure noise. With 12 seeds the same window gives modular $\times 1.36$ vs monolith $\times 2.09$, and the full six-size slopes separate at $p = 3.3 \times 10^{-4}$. Two-point scaling claims on a heavy-tailed init quantity are not claims; this is why §3.2 leads with the exact structural identity and reports the growth claim only with 12 seeds and six sizes.

E The mass-lr inducer: full grid (CM-5; N7)

3 seeds, 1500 ep, uniform init; `align=Spearman(learned  $M$ , true band)`, `spread=std(log  $M$ )`.

N	ratio	mult	align	log-spread	eval-MSE
2	4×	3	+1.00	0.263	0.274
2	4×	10	+1.00	0.553	0.033
2	4×	30	+1.00	0.767	0.022
2	4×	100	+1.00	1.171	0.373
2	16×	3	+1.00	0.386	1.372
2	16×	10	+1.00	0.488	0.633
2	16×	30	+0.33	0.071	0.366
2	16×	100	−1.00	0.176	0.332
4	4×	3	+1.00	0.222	0.149
4	4×	10	+1.00	0.442	0.028
4	4×	30	+1.00	0.646	0.067
4	4×	100	+1.00	1.718	35.27 ± 47.26
4	16×	3	+0.93	0.347	0.680
4	16×	10	+0.80	0.611	0.296
4	16×	30	+0.20	0.678	0.197
4	16×	100	−0.20	1.459	0.800

N7: the “ ~ 0.08 ceiling / optimizer never finds it” reading is a 300-epoch budget artifact — at 1500 ep the uniform lattice orders masses correctly 5/5 seeds (Spearman +0.89) and a 10× mass-lr amplifies spread to 0.52; but the designed magnitude is never reached (designed fits 7–14× better). Ordering inducible; magnitude designed.

F Recovery ladder and loan curve (CM-1; C-9)

F.1 Loan curve (cumulative MSE, 3 seeds, fit-gap-anatomy item 2):

horizon	twin	chlu	broken_vol	LSTM
10	2.9×10^{-6}	2.2×10^{-5}	1.1×10^{-4}	7.5×10^{-2}
100	2.4×10^{-4}	1.2×10^{-2}	4.1×10^{-3}	4.9×10^{-2}
500	1.1×10^{-2}	1.8×10^{-1}	9.7×10^{-2}	7.7×10^{-2}
1000	3.2×10^{-1}	2.0×10^{-1}	1.2×10^{-1}	9.9×10^{-2}
5000	1.96×10^2	2.2×10^{-1}	1.4×10^{-1}	1.3×10^{-1}

Loan called between 500 and 1000 (crosses ≈ 700). CLU is not the lowest plateau (broken-vol 0.14, LSTM $0.13 < \text{CLU } 0.22$); the asset is boundedness by construction. **F.2 Recovery ladder (fit-gap-anatomy item 3; twin gap 0.202):** twin 0.0047 (100%); CLU $\gamma=0$ 0.2066 (0%, BIBO 1.00, drift 0.778, $\mu^2 5.2 \times 10^{-3}$); $+\gamma$ **global** 0.0216 (**92%, all guarantees preserved**); $+\gamma_\varphi$ 0.2548 (−24%, wrong lever). Reach rungs certificate-only: $+S^{(M)} \det = 1$ (3×10^{-7}), inj $2.28 \leq e^2$, $\Delta q = 2.18$; +inter-unit wormhole $\kappa=0$ bit-exact reduction re-verified.

G Prominent negatives registry (C-9)

N7 designed-not-emergent (ordering *is* inducible) — App E. **N8** 100× mass-lr inverts ordering — App B.3. **N9** slow-first curriculum did not help — App B.3. **N12** governor + friction field do not compose (deploy the field alone). **N16** trained-coupling exponent was read “inconclusive” — **now RESOLVED/re-attributed** (artifact of the *learned* random- W $U(1)$ breaking, not physics; on `channel_spring` / designed $SO(2)$ units the exponent is verified flat across $N \leq 16$, $-0.49 \pm 0.02 / -0.91 \pm 0.03$, pre-registered) — §3.3, Figs. 3–7, App C. **N22** friction never stabilizes a saddle;

governor blind to isoenergetic escape (theory note §5) — V3 assumes the licensed-contraction forgetting column. **Superseded internal measurement (process negative; App D.1):** a 3-seed, 2-point ($N \in \{4, 8\}$) reading of the interference growth suggested equal slopes ($\times 2.56$ vs $\times 2.18$); the per-seed 95% CI on the modular factor was ± 2.27 — the comparison carried no information. The 12-seed six-size measurement reverses it ($\times 1.36$ vs $\times 2.09$; slopes diverge, $p = 3.3 \times 10^{-4}$). **The firewall is not a physics result (App H):** a block-diagonal potential with separate per-block parameters has $S \equiv 0$ *exactly* — a strictly better firewall than our lattice’s $\bar{R}_{\text{edge}} \approx 6.5 \times 10^{-5}$, with no Hamiltonian, no symplectic structure and no coupling graph. Nothing physics-specific buys interference confinement; what the lattice buys is a *priced channel* through the firewall.

H The block-monolith and coordination-number controls (§3.2(i),(iii))

(Source: *v3-interference-extra*; flag-provenance A.2b. 12 seeds (block/scaling), 8 seeds (topology), $N \leq 16$, measured at initialization.)

H.1 Full five-arm block-monolith table. A single potential $V : \mathbb{R}^{Nd} \rightarrow \mathbb{R}$ (one module; no lattice, no coupling graph, no Hamiltonian coupling) with an explicitly block-diagonal Hessian ($\partial^2 V / \partial q_i \partial q_j = 0$, $i \neq j$), in three parameter-sharing regimes, against the modular lattice and the naive monolith.

arm	params ($N=2 \rightarrow 16$)	$S(2)$	$S(4)$	$S(8)$	$S(16)$	per-pair $\bar{R}$	b over $[2, 16]$
block-untied sep. θ_i	2,370 $\rightarrow$ 18,960	0.0	0.0	0.0	0.0	0.0	undef. ($S \equiv 0$)
modular ($\kappa_c=0.05$)	2,382 $\rightarrow$ 19,112	6.4e−5	9.8e−5	1.3e−4	1.1e−4	$\approx 6.5\text{e}−5$	$+0.46 \pm 0.31$
block-trunk	1,218 $\rightarrow$ 1,680	7.08e−2	2.20e−1	4.88e−1	1.05	0.065	$+1.51 \pm 0.36$
monolith	1,253 $\rightarrow$ 2,177	2.18e−1	6.40e−1	1.34	2.32	0.19	$+1.18 \pm 0.17$
block-tied (deep-sets)	1,185 (const.)	9.64e−1	2.90	6.78	1.45×10^1	0.97	$+1.29 \pm 0.01$

Exact-zero audit over all seeds/sizes (max off-diagonal R ; nonzero far-field entries): modular 9.34×10^{-4} / **0** of 4,656; **block_untied** 0.000000×10^0 / **0**; monolith 6.22×10^{-1} / 4,656; **block_trunk** 3.50×10^{-1} / 4,656; **block_tied** 1.27 / 4,656. Diagonal $R[A, A] = 1.000000$ in all 72 runs of every arm, so the **block_untied** zero is genuine, not a degenerate normalization of a dead self-response. Readings: (a) *block structure alone recovers nothing* (**block_trunk** shares only a trunk and still leaks, $\bar{R} = 0.065$, $b = +1.51$); (b) *block structure + disjoint parameters is a perfect firewall* ($S \equiv 0$); (c) *interference is not capacity* — the 1,185-parameter tied arm (constant in N) is worst, the 18,960-parameter untied arm exactly zero, and **block_tied**’s $S = 14.5$ at $N = 16$ is $129,033\times$ the modular value and $6.2\times$ the naive monolith; (d) the modular lattice equals **block_untied** at $\kappa_c = 0$ (measured 0.0), so its $\approx 6.5 \times 10^{-5}$ leak is precisely the price of having a coupling graph.

H.2 Coordination-number control (chain/ring/circulant-4). 8 seeds; the identity $S = \deg \cdot \bar{R}_{\text{edge}}$ holds to $S/(\deg \cdot \bar{R}_{\text{edge}}) = 1.000000$ in *every* cell:

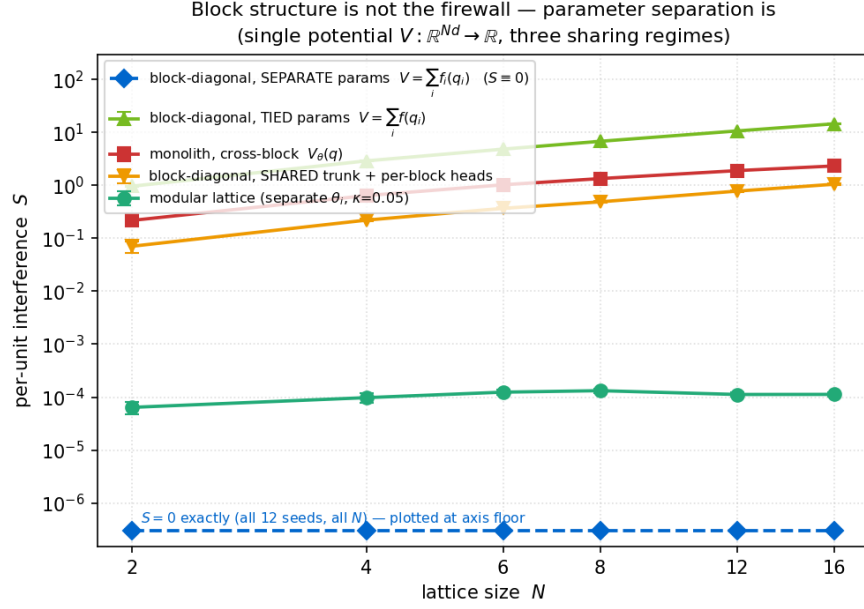


Figure 9: [evidence] **Parameter separation, not block structure, buys the firewall.** Per-unit received interference S vs N for five arms (12 seeds; log- y). `block_untied` is drawn at the axis floor with an explicit “ $S = 0$ exactly” annotation; `block_tied` (deep-sets, 1,185 parameters constant in N) is the worst arm, above the naive monolith (`v3-interference-extra`).

topology	N	mean degree	S	$\bar{R}_{\text{edge}}$	$S/(\text{deg} \cdot \bar{R}_{\text{edge}})$	non-edge nonzero
chain	4	1.50	$7.934 \times 10^{-5} \pm 3 \times 10^{-5}$	5.289×10^{-5}	1.000000	0
chain	8	1.75	$1.496 \times 10^{-4} \pm 3 \times 10^{-5}$	8.549×10^{-5}	1.000000	0
chain	16	1.88	$1.148 \times 10^{-4} \pm 4 \times 10^{-5}$	6.124×10^{-5}	1.000000	0
ring	4	2.00	$1.189 \times 10^{-4} \pm 4 \times 10^{-5}$	5.944×10^{-5}	1.000000	0
ring	8	2.00	$1.724 \times 10^{-4} \pm 5 \times 10^{-5}$	8.622×10^{-5}	1.000000	0
ring	16	2.00	$1.315 \times 10^{-4} \pm 5 \times 10^{-5}$	6.575×10^{-5}	1.000000	0
circulant-4	8	4.00	$2.829 \times 10^{-4} \pm 8 \times 10^{-5}$	7.072×10^{-5}	1.000000	0
circulant-4	16	4.00	$2.642 \times 10^{-4} \pm 4 \times 10^{-5}$	6.605×10^{-5}	1.000000	0

Slopes: chain $+0.264 \pm 0.171$ ($p = 0.008$, a real end-effect from the degree ramp), ring $+0.071 \pm 0.183$ ($p = 0.391$, flat), circulant-4 -0.066 ± 0.315 ($p = 0.636$, flat). Ring→circulant-4 scales S by $2.01\times$ at $N = 16$ ($1.64\times$ at $N = 8$); the shortfall at $N = 8$ is $\bar{R}_{\text{edge}}$ init scatter ($\pm 30\%$ across lattices), **not** a degree effect — the identity is exact per-run, so we quote the identity and never “ $2.0\times$ ” as a measured constant.

H.3 Banding orthogonality, exact. $\max |R_{\text{banded}} - R_{\text{uniform}}| = 0.000 \times 10^0$ over 6 sizes $\times$ 12 seeds: the banded and uniform interference kernels are **bit-identical**, upgrading the earlier “equal to every printed digit.” V_θ has no `log_mass` dependence; banding and the firewall are orthogonal knobs.

References (hermetic; C-8)

Open bibliography slots (v0.4): three strings remain unresolved and are marked, never fabricated — (a) Mo (2026); (b) Di Bernardo et al./Keller; (c) the checkpointing $O(\sqrt{T})$ + RevNet/momentum-network anchors

Interference is bounded by coordination number, not width ($S = \deg \cdot \bar{R}_{\text{edge}}$ exactly; $R \equiv 0$ off the coupling graph)

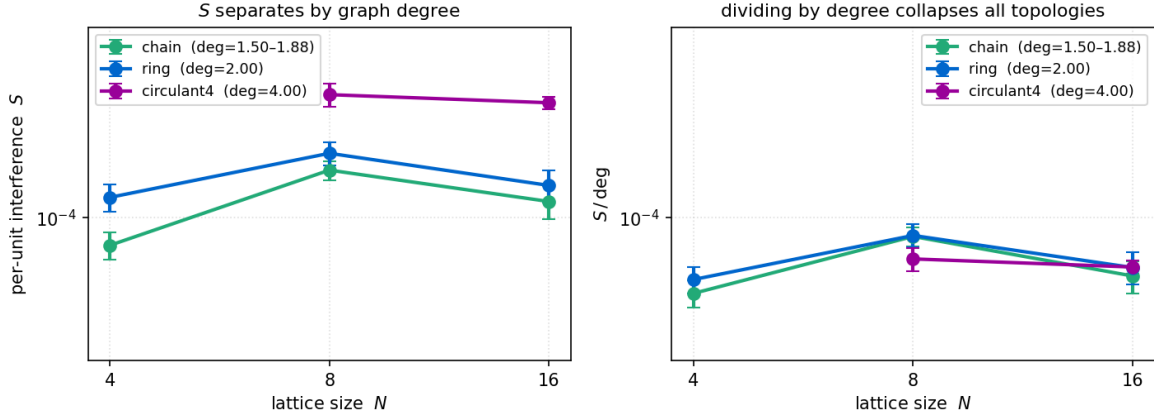


Figure 10: **[evidence] Interference is bounded by coordination number, not width.** (left) S vs N separates by graph degree (chain 1.50–1.88, ring 2.00, circulant-4 4.00); (right) dividing by degree collapses all three topologies. $R \equiv 0$ off the coupling graph in every topology — including the circulant, where graph distance and index distance disagree. 8 seeds, laptop-CPU (v3-interference-extra).

of the §4 reversible paragraph. They were scheduled to land with the venue scout’s bibliography sub-task, whose report does not exist at this revision.

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